

6-DOF Robot Arm Sorting Cubes

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Introduction

One of the most common tasks for robotics is detecting and moving objects. Some examples include household robots, folding clothes and moving objects. In this project, the mission will be simplified to sorting cubes with different colors and putting them into their bins respectively.

Why and how relevant to CSCI 4551

Our project is relevant to CSCI 4551 because it applies the main robotics pipeline studied in the course: sensing, processing, planning, and acting in a simulated robot system. In CSCI 4551, we mainly learned manipulating ROS2-based robot systems built from connected sensors, actuators, nodes, publish/subscribe communication, and visualization tools such as RViz. Our project follows this Sense-Think_Act structure: An overhead Depth camera will be used to capture color and depth frames of the workspace, identifying each object and estimating its 3D position and orientation on the table. Then, the system classifies each object by color, selects the matching bin, computes a grasp pose from the depth data and plans a collision-free trajectory through pre-grasp, grasp, lift, transport, and drop. Finally in the Act stage, a 6-DOF arm in Gazebo executes the planned trajectory, closes the gripper on the object, moves it to the correct bin, and releases. MoveIt is used but it is not the main focus of this project.

Methodology

Generally we built a pick-and-place system in Gazebo based using the ros2 architecture. The goal is to detect different colored cubes on a table (spawned at random positions and

orientations), determine which bin each cube belongs to, then control a 6-DOF robotic arm to pick up each cube and drop it into its bin. The first part we implemented was the perception node. We used OpenCV to segment each color channel and scan them for contours to find the centroids of each cube. The camera data from the simulated camera is used to identify cubes on the table, their positions, and orientations. In order to help the robot arm understand the cube locations relative to the table and robot base, the detected pixel coordinates need to be converted into world-frame coordinates.

The next thing we focused on is the controller. This node subscribes to the detection of cubes. Once it receives something, it determines the pick-and-place order. The controller will then match each cube's color to the corresponding bin color and generate the target poses for later `pick_and_place`. For each cube, the controller will send a target pose to the motion-planning system, first is the grasp position above the cube and then is the release position above the bin.

The final mission for us is motion planning and collision avoidance. MoveIt will later be used to plan collision-free trajectories for the 6-DOF robotic arm, but we first need to test if the arm can follow a simple trajectory. For each pick-and-place action, the arm will follow this sequence: a pre-grasp pose above the cube, a pose down to the cube, closing gripper and lifting, a motion toward the correct bin, and opening grippers.

Distribution of work:

- Shawn: Setting up the ros2 system, Gazebo simulation environment, and spawning the different colored cubes. Finally implementation of the perception node
- Stanley: Subscribing to the `detected_cubes` and fixed bin positions and generating motion, mainly focusing on the arm manipulation, the `pick_and_place_node`.

Conclusion

In short, this project demonstrates a simplified pipeline trying for robotic pick-and-place manipulation using OpenCV, ROS2, Gazebo simulation, and MoveIt. It shows a complete structure that connects perception and action by detecting colored cube poses in the workspace,

getting their positions and orientations, sorting each cube to their matching colored bin, and manipulating a 6-DOF robotic arm to complete the sorting mission. The current stage of this project is complete aside from the grasping and transportation of cubes using the 6DOF arm. Currently, the robot arm can only move above each cube in Gazebo and complete the movement to the bin but is not able to perform the grasp. Next steps would be to tune the arms grasping profile, as it has all the parts needed to function, the IK solver is working, however things just are not lining up quite right.